

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 6: Class A Take-Off Distances & Balanced Field Length

Syllabus Source: CAE Oxford (pp. 195-238) | Pass Target: >= 90% AviationExam

1. DECLARED DISTANCES ARCHITECTURE

Airport authorities declare four official runway distances that establish the take-off and landing boundaries:

Declared Distance Code	Physical Runway Components Included	Regulatory Meaning & Usage
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TORA (Take-Off Run Available)	Paved physical runway length suitable for normal ground run.	Baseline distance for all wheel acceleration.
TODA (Take-Off Distance Available)	TORA + Clearway (unobstructed water/grass area, max 50% TORA).	Total distance to accelerate and climb to 35 ft screen height.
ASDA (Acc-Stop Distance Available)	TORA + Stopway (load-bearing overrun strip suitable for abort).	Total distance available to accelerate to V1 and stop completely.
LDA (Landing Distance Available)	Paved runway from threshold to end of roll.	Distance available for landing touch down and rollout.

2. THE BALANCED FIELD LENGTH CONCEPT

A runway is considered a 'Balanced Field' when: TODA equals ASDA (Clearway = Stopway).

- **Balanced V1:** The unique decision speed where Accelerate-Stop Distance (reject at V1) EXACTLY EQUALS the One-Engine-Inoperative Take-off Distance (continue at V1 to 35 ft screen height)!
- **Variables that SHIFT V1:** Higher Mass -> Increases V1. Higher Flap Setting -> Decreases V1. Uphill Slope -> Increases V1. Headwind -> Decreases groundspeed at V1. Contaminated Runway (water/slush) -> DRATICALLY REDUCES V1 (because braking friction is degraded, you must abort earlier)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Declared Distances & Clearway L

Declared Distance Formulas: - TORA = Take-Off Run Available (paved surface). - ASDA = TORA + Stopway. - TODA = TORA + Clearway (legal limit: clearway cannot exceed 50% of TORA!). Balanced Field Length Concept: - Occurs when Accelerate-Stop Distance Required equals Accelerate-Go Distance Required with engine failure at V1. - Balances field length to minimize required runway and maximize allowable takeoff mass.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Declared Distances & Clearway Limits

SCENARIO (Standard EASA Airport Planning Drill): An airport publishes the following physical runway survey data: - Paved Runway length = 2,400 meters - Stopway paved overrun strip = 150 meters - Clearway unobstructed obstacle-free area = 1,400 meters

QUESTION: What are the legal declared distances for TORA, ASDA, and TODA under EASA CS-25 rules? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate TORA (Take-Off Run Available): - TORA is strictly the physical paved runway length = 2,400 meters. Step 2: Calculate ASDA (Accelerate-Stop Distance Available): - ASDA = TORA + Stopway - ASDA = 2,400 m + 150 m = 2,550 meters. Step 3: Calculate TODA (Take-Off Distance Available) with Clearway Limit Check: - Rule: Clearway credit CANNOT EXCEED 50% of TORA! - Maximum allowable clearway = $0.50 \times 2,400 \text{ m} = 1,200 \text{ meters}$. - The physical clearway is 1,400 m, but we can ONLY LEGALLY CREDIT 1,200 meters! - Legal TODA = TORA + Allowed Clearway = 2,400 m + 1,200 m = 3,600 meters! FINAL ANSWER: TORA = 2,400 m; ASDA = 2,550 m; TODA = 3,600 m (limited by the 50% clearway rule!).

[!] EXAM TRAP (AVIATIONEXAM) - The Clearway 50% TORA Rule Trap

- Even if an airport has 3 kilometers of open sea beyond the runway, you CANNOT add more than 50% of TORA to TODA! - E.g. If TORA = 2,000 m and Clearway = 1,500 m, TODA is NOT 3,500 m! TODA is capped at $2,000 + 1,000 = 3,000 \text{ m}$!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the maximum clearway length that can legally be included in Take-Off Distance Available (TODA)?**

- [A]: Unlimited
- [B]: Not more than 50% of TORA
- [C]: Equal to the stopway length
- [D]: 1,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CS-25 and ICAO Annex 14, the length of clearway that can be credited to determine TODA must not exceed half the length of TORA.

Q2: In a 'Balanced Field Length' take-off calculation, which two distances are equal?

- [A]: TORA and LDA
- [B]: The Accelerate-Stop Distance and the One-Engine-Inoperative Take-Off Distance to 35 ft
- [C]: Ground roll and climb distance
- [D]: Clearway and runway length

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A balanced field exists when the distance required to accelerate to V1 and stop equals the distance required to accelerate, lose an engine at V1, and reach screen height.

Q3: How does CONTAMINATION (e.g. 5 mm standing water) affect the optimal V1 decision speed?

- [A]: Increases V1

- [B]: SUBSTANTIALLY DECREASES V1, because degraded wheel braking requires aborting earlier, and remaining runway allows continued take-off to a reduced 15 ft screen height
- [C]: Has zero effect
- [D]: Eliminates V1

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. On contaminated runways, braking friction is severely reduced. V1 must be reduced to allow stopping within ASDA if an abort occurs; screen height for continue is lowered to 15 ft.